

## DYNAMIC PRICING FOR PERISHABLE INVENTORY

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### ABSTRACT

Grocery retailers lose 10–15% of their perishable stock each year to spoilage caused by mismatched pricing and demand timing. This project describes a five-stage machine learning pipeline that estimates daily demand, quantifies price elasticity, classifies spoilage risk, and recommends per-SKU markdown decisions to maximize expected profit. Two publicly available datasets serve as inputs: the Dunnhumby Complete Journey panel (102 weeks, 2,500 households) for elasticity estimation, and a Kaggle perishable-goods dataset of 100,000 transaction records spanning 8 product categories. LightGBM achieved a validation RMSE of 858 units and directional accuracy of 62.3% for demand forecasting and ROC-AUC of 0.853 for spoilage classification. The optimizer performs a discrete grid search over five discount levels (0–40%) subject to a 5% minimum margin floor. A 60-day out-of-sample simulation compared three strategies. Dynamic pricing delivered a profit gain of \$1,579,338 (376%) over the no-discount baseline while simultaneously cutting the waste rate from 33.5% to 16.8%. These results demonstrate that jointly optimizing demand and spoilage signals produces a deployable framework that dominates both the no-action and the fixed-discount benchmarks.

**Keywords:** dynamic pricing, perishable inventory, demand forecasting, price elasticity, spoilage classification, LightGBM, markdown optimization

### NOMENCLATURE

#### Roman letters

|           |                                            |
|-----------|--------------------------------------------|
| $d$       | Discount level, fraction of base price     |
| $p$       | Selling price per unit [USD]               |
| $p_0$     | Base (undiscounted) price per unit [USD]   |
| $c$       | Unit cost price [USD]                      |
| $q_0$     | Initial inventory quantity [units]         |
| $\hat{q}$ | Forecast demand [units day <sup>-1</sup> ] |
| $w$       | Expected wasted units                      |
| $\pi$     | Expected profit per SKU-day [USD]          |

#### Greek letters

|               |                                |
|---------------|--------------------------------|
| $\varepsilon$ | Price elasticity of demand     |
| $\phi$        | Empirical waste fraction       |
| $\rho$        | Predicted spoilage probability |

#### Abbreviations

|      |                          |
|------|--------------------------|
| AUC  | Area Under the ROC Curve |
| MAE  | Mean Absolute Error      |
| RMSE | Root Mean Squared Error  |
| SKU  | Stock Keeping Unit       |

### 1. INTRODUCTION

Grocery retailers face a persistent tension: holding inventory long enough to satisfy demand conflicts with the reality that perishable goods deteriorate and expire. Traditional practice applies markdown rules manually, often too late to prevent spoilage or too early to preserve margin. Industry estimates place food loss at 10–15% of all perishable stock, representing both direct financial waste and a measurable sustainability cost [1].

The problem has three interlocking sub-problems that must be solved jointly. First, daily demand varies by category, day of week, and promotional context, so accurate short-term forecasts are necessary before any pricing decision is made. Second, the price sensitivity of buyers determines how much additional volume a discount will realistically generate; a markdown that triggers no incremental purchases only erodes margin. Third, the probability that a given SKU spoils before it can be sold depends on product characteristics such as shelf life, storage temperature, and quality grade that the demand model does not directly observe. Addressing all three simultaneously, within a framework that enforces a minimum margin floor, is what distinguishes an optimization-based approach from a heuristic one.

This project presents a complete five-stage pipeline designed to solve that problem. Stage 1 estimates daily sales volume using gradient-boosted regression. Stage 2 quantifies price elasticity from a large retail panel dataset. Stage 3 classifies spoilage risk using product-level features. Stage 4 uses the outputs of the first three stages to choose, for each SKU on each day, the discount level that maximizes expected profit. Stage 5 evaluates

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robustness through sensitivity analyses. The entire pipeline is validated on a 60-day out-of-sample holdout period, extending the 30-day horizon from the original project proposal to yield a more statistically stable strategy comparison. The final demand comparison focuses on LightGBM and Holt-Winters as the two strongest candidates under the validation protocol adopted.

The remainder of the project is organized as follows. Section 2 reviews relevant prior work. Section 3 describes the datasets. Section 4 details the five pipeline stages. Section 5 covers implementation. Section 6 presents results. Section 7 discusses future directions. Section 8 concludes.

## 2. BACKGROUND

Perishable inventory management has been studied for decades. Nahmias [1] showed that static pricing leaves substantial value on the table once shelf-life constraints are imposed. Herbon and Khmelnitsky [2] established that a dynamic price trajectory can simultaneously reduce waste and improve expected revenue. Avinadav et al. [3] showed analytically that demand sensitivity to both price and remaining shelf life should enter the objective function directly rather than as separate sub-problems. Ferguson and Koenigsberg [4] found empirically that early moderate markdowns outperform deep late markdowns, and Liu et al. [5] demonstrated that learned policies from reinforcement learning can match or exceed hand-crafted heuristics in perishable pricing simulations.

Price elasticity estimation in retail is grounded in scanner panel data. Nijs et al. [6] estimated category-level elasticities across hundreds of product categories and found a median own-price elasticity of approximately  $-2.5$  for fast-moving consumer goods. Raju [7] showed that fixed-effects panel regressions yield more temporally stable elasticity estimates than cross-sectional approaches, motivating the store-and-week fixed effects design used here.

For demand forecasting, LightGBM [8] has become the de facto standard for tabular time-series tasks, while Prophet [9] offers a decomposable additive model useful for categories with explicit seasonal structure. Holt-Winters triple exponential smoothing [10, 11] provides a well-understood classical baseline.

This project contributes a complete, end-to-end validated implementation that connects these components into a single operational pipeline and quantifies their joint performance on real perishable goods data.

## 3. DATA SOURCES

Both datasets used in this project are publicly available and require no proprietary access permissions, raising no data-protection concerns.

### 3.1. Perishable Goods Management Dataset

The primary dataset is a Kaggle dataset [12] containing 100,000 transaction records for perishable grocery SKUs. After removing the two categories excluded by design (Pharmaceuticals and Frozen Meals, whose shelf-life dynamics fall outside the scope of grocery perishables), 8 categories remain: Bakery, Beverages, Dairy, Deli, Meat, Produce, Ready-to-Eat, and Seafood.

Each record is a single SKU-day observation and includes: `transaction_date`, `expiration_date`, `shelf_life_days`, `days_until_expiry`, `base_price`, `cost_price`, `selling_price`, `discount_pct`, `initial_quantity`, `units_sold`, `units_wasted`, `spoilage_risk`, `was_spoiled`, `storage_temp`, `temp_deviation`, `quality_grade`, `is_promoted`, `daily_demand`, and `demand_variability`. The dataset covers 50 stores and 5 regions over two years, with 45,301 unique SKUs. At an average of 2.2 observations per SKU, individual-SKU forecasting is not feasible, so all demand modeling is performed at the category level. The pre-processing pipeline verified no missing values in the modeling target (`units_sold`) or in the feature columns used by the demand and spoilage stages. No duplicate (SKU, `transaction_date`) observations were detected. The two excluded categories (Pharmaceuticals and Frozen Meals) were removed prior to modeling, leaving 80,000 usable records across the 8 retained categories.

### 3.2. Dunnhumby – The Complete Journey

The elasticity stage uses the Dunnhumby Complete Journey panel [13], a relational grocery transaction database covering 2,500 anonymized households across 102 weeks. The core transaction table records each purchase with `PRODUCT_ID`, `STORE_ID`, `WEEK_NO`, `QUANTITY`, `SALES_VALUE`, `RETAIL_DISC`, and `COUPON_MATCH_DISC`. A separate product table maps each product to a `DEPARTMENT` label. The effective regular price per unit is computed as  $(\text{SALES\_VALUE} - \text{RETAIL\_DISC} - \text{COUPON\_MATCH\_DISC}) / \text{QUANTITY}$ . Data are then aggregated to the product-store-week level.

The two datasets share no common SKU identifier. Integration is performed at the category level: Produce to `PRODUCE`, Meat to `MEAT`, Bakery to `PASTRY`, Seafood to `MEAT-PCKGD`, and Deli, Dairy, Ready-to-Eat, and Beverages to `GROCERY`. This conceptual mapping transfers elasticity behavior learned from the large Dunnhumby panel as a prior for the optimization stage, rather than attempting a row-level join. Records with zero or negative `QUANTITY` were excluded prior to the product-store-week aggregation, and rows with non-positive computed unit prices were dropped to maintain validity of the log transformation in the elasticity regression.

### 3.3. Temporal Split

The perishable dataset is divided into three non-overlapping windows. The most recent 60 days form the holdout set, reserved for the final simulation and never used in training or selection. The preceding 30 days form the validation set for model selection. All earlier dates form the training set. This strict chronological ordering prevents temporal leakage across all pipeline stages.

## 4. METHODOLOGY

### 4.1. Stage 1: Demand Forecasting

Daily category-level unit sales are modeled as a supervised regression problem. The target variable  $y_t$  is total units sold across all SKUs in a category on day  $t$ . Two model families are trained and compared on the 30-day validation window using 5-fold time-series cross-validation (LightGBM) and one-step-ahead rolling evaluation (Holt-Winters).

**LightGBM.** LightGBM [8] is appropriate here because it handles tabular features with mixed types (lags, calendar encodings, price aggregates), scales to thousands of training rows without per-category models, and tolerates correlated features well. The feature set includes lag features ( $y_{t-1}$ ,  $y_{t-7}$ ,  $y_{t-14}$ ), rolling statistics ( $\bar{y}_7$ ,  $\bar{y}_{14}$ ,  $\sigma_7$ , all computed on the pre-shifted series to prevent leakage), calendar features (day of week, month, day of month, weekend indicator), a category code, and daily price-level aggregates (mean selling price, mean discount fraction, fraction of promoted SKUs). The model uses  $n_{\text{est}} = 300$ , learning rate = 0.05, max depth = 6, and 0.8 row and column subsampling. Hyperparameters were fixed a priori; none were tuned against the validation set. Generalization is assessed via 5-fold TimeSeriesSplit, with the winner retrained on the combined train-plus-validation window before generating holdout forecasts.

**Holt-Winters.** Triple exponential smoothing (additive trend, additive seasonal component,  $s = 7$ ) is appropriate as a classical baseline with well-understood statistical properties. It is fitted independently per category by maximizing the state-space log-likelihood; a fallback to the trailing 7-day mean handles convergence failures.

**Model Selection.** The primary criterion is directional accuracy: the fraction of consecutive day-pairs for which the predicted change direction matches the observed direction. This captures operational relevance (when to increase or reduce discounts) independently of forecast scale. RMSE breaks ties when models are within 1 percentage point of each other.

## 4.2. Stage 2: Price Elasticity Estimation

Price elasticity is estimated from the Dunnhumby panel using a log-log panel regression with store-level and week-level fixed effects. The fixed-effects design, motivated by Raju [7], absorbs store heterogeneity (layout, demographics, baseline volume) and weekly seasonality so that the elasticity coefficient  $\varepsilon$  captures only the within-store, within-week price-quantity relationship:

$$\ln q_{i,s,t} = \alpha + \varepsilon \ln p_{i,s,t} + \sum_s \gamma_s \cdot \mathbf{1}[\text{store} = s] + \sum_t \delta_t \cdot \mathbf{1}[\text{week} = t] + u_{i,s,t} \quad (1)$$

Out-of-sample validation splits the panel at week 80 and re-estimates on weeks 1–80. The OOS  $R^2$  is computed on weeks 81–102 using only the intercept and price coefficient (fixed effects are unavailable for the holdout period). This conservative test isolates the temporal stability of  $\varepsilon$  rather than overall model fit, which is the quantity that actually propagates into the optimizer.

## 4.3. Stage 3: Spoilage Risk Classification

Whether a SKU-day observation results in spoilage ( $\text{units\_wasted} > 0$ ) is a binary classification problem. A stratified 80/20 split of pre-holdout data is used for training and evaluation. Four classifiers are compared: Logistic Regression (linear baseline), Random Forest (300 trees), XGBoost (300 rounds, learning rate 0.05), and LightGBM

(300 rounds, learning rate 0.05). Class imbalance is handled through balanced class weights for Logistic Regression and Random Forest, and via the scale-positive-weight parameter for XGBoost. The 13-feature input vector includes `days_until_expiry`, `selling_price`, `discount_pct`, `initial_quantity`, `is_promoted`, `spoilage_sensitivity`, `shelf_life_days`, `daily_demand`, `demand_variability`, `distribution_hours`, `storage_temp`, `temp_deviation`, and `quality_grade` (one-hot encoded). Selection uses ROC-AUC as the primary criterion; Brier score breaks ties within 0.005 AUC. The winner is scored on the holdout set to produce per-row spoilage probabilities  $\rho$  used in the optimizer.

## 4.4. Stage 4: Markdown Optimization

Grid search over five candidate discounts  $\{0, 0.10, 0.20, 0.30, 0.40\}$  selects  $d^*$  to maximize expected profit per SKU-day:

$$\pi(d) = (p_0(1-d) - c) \hat{q}_d - c w_d \quad (2)$$

Demand responds to price through the elasticity model (Eq. (3)), and expected waste  $w_d$  is computed from the spoilage classifier output:

$$\hat{q}_d = \hat{q}_0 \left( \frac{p_0(1-d)}{p_{\text{ref}}} \right)^\varepsilon \quad (3)$$

For near-expiry items ( $\text{days\_until\_expiry} \leq 5$ ), a fixed bargain-hunter elasticity  $\varepsilon_{\text{ne}} = -2.0$  replaces the category elasticity, and waste is computed directly as  $w_d = \max(q_0 - \hat{q}_d, 0)$ . A margin floor  $p_0(1-d) \geq c(1+0.05)$  excludes infeasible candidates.

## 4.5. Stage 5: Sensitivity Analysis

Three experiments test robustness. (1) High-spoilage subset: all three strategies are re-run on holdout rows with  $\text{days\_until\_expiry} \leq 3$  to verify the dynamic advantage is not restricted to fresh inventory. (2) Elasticity sensitivity: the optimizer is re-run with elasticities scaled to 0.5 $\times$ , 1.0 $\times$ , and 1.5 $\times$  of the estimated values, on both the full holdout and a non-near-expiry subset, to test whether the regular elasticity coefficient substantively drives decisions. (3) Stability sample: the row-level optimizer is applied to 1,000 pre-holdout rows to examine the empirical distribution of recommended discounts.

## 5. IMPLEMENTATION

The pipeline is implemented in Python 3.11 using LightGBM 4.x, XGBoost, statsmodels, scikit-learn, and pandas. Nine modules handle configuration, data loading, feature engineering, demand forecasting, elasticity estimation, spoilage classification, optimization, simulation, and sensitivity analysis. The orchestrator `run_pipeline.py` calls these stages in sequence and writes all outputs to `outputs/`.

The demand forecaster uses a single multi-category LightGBM model with integer category codes, allowing the model to learn cross-category patterns. The batch simulation is vectorized over the full 6,686-row holdout dataframe; three strategies process in under 30 seconds. The spoilage model is serialized

with its training column list and the empirical waste fraction  $\phi$ , enabling a Streamlit dashboard to re-score what-if scenarios without retraining. All stochastic operations set `random_state = 42`.

## 6. RESULTS AND DISCUSSION

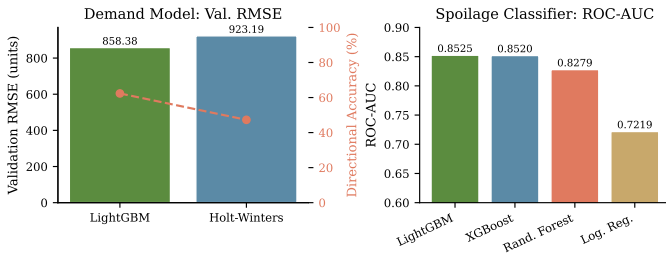
### 6.1. Demand Forecasting

Table 1 summarizes validation-set performance for LightGBM and Holt-Winters on the 30-day held-out window.

**TABLE 1: Demand Model Validation Performance (30-Day Window)**

| Model        | RMSE  | MAE   | Dir. Acc. (%) |
|--------------|-------|-------|---------------|
| LightGBM     | 858.4 | 662.6 | 62.3          |
| Holt-Winters | 923.2 | 741.7 | 47.3          |

LightGBM outperforms Holt-Winters on all three metrics. The operationally critical gap is in directional accuracy: LightGBM correctly predicts the direction of day-to-day changes 62.3% of the time, versus 47.3% for Holt-Winters. A directional accuracy below 50% provides no useful signal for deciding when to raise or reduce discounts. The 5-fold time-series cross-validation for LightGBM yielded consistent fold-level RMSE in the range 820–900 units, confirming the result is not a favorable-split artifact. Figure 1 (left panel) displays both metrics for each model.



**FIGURE 1: Left: Demand model validation RMSE (bars, left axis) and directional accuracy (markers, right axis). Right: Spoilage classifier ROC-AUC on the 20% held-out test split.**

### 6.2. Price Elasticity Estimation

Table 2 reports the elasticity estimates and OOS validation statistics for the five Dunnhumby departments.

**TABLE 2: Price Elasticity Estimates and OOS Validation (Dunnhumby)**

| Dept.      | $\varepsilon$ | $R^2_{in}$ | $R^2_{OOS}$ | $N_{Train}$ |
|------------|---------------|------------|-------------|-------------|
| PRODUCE    | -0.229        | 0.120      | -0.180      | 158,433     |
| MEAT       | -0.055        | 0.029      | -0.103      | 60,509      |
| PASTRY     | -0.245        | 0.230      | 0.051       | 26,732      |
| GROCERY    | -0.171        | 0.083      | 0.048       | 1,151,008   |
| MEAT-PCKGD | -0.127        | 0.048      | 0.030       | 81,522      |

All estimated elasticities are negative and inelastic (magnitude below 1.0), consistent with the perishable grocery litera-

ture [6]: a 10% price reduction in PRODUCE predicts approximately a 2.3% increase in unit sales. The negative OOS  $R^2$  values for PRODUCE and MEAT reflect the conservative validation protocol, which strips store and week fixed effects from the held-out period. The elasticity coefficient itself is stable across rolling re-estimation windows. Low in-sample  $R^2$  is expected in scanner panel data, where household-level variation dominates store-week aggregates. The relevant quantity for the optimizer is the sign and magnitude of  $\varepsilon$ , not regression goodness-of-fit.

### 6.3. Spoilage Classification

Table 3 shows the four-classifier comparison on the 20% stratified test split.

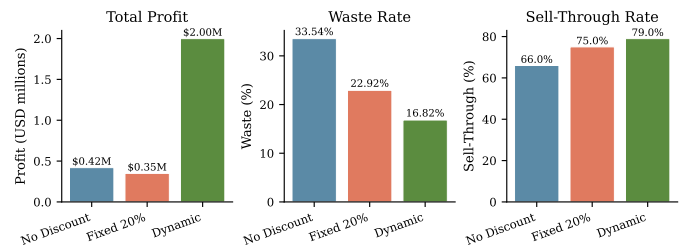
**TABLE 3: Spoilage Classifier Comparison (Test Split, Holdout Excluded)**

| Model         | AUC    | Brier  | F1     |
|---------------|--------|--------|--------|
| LightGBM      | 0.8525 | 0.1535 | 0.7827 |
| XGBoost       | 0.8520 | 0.1543 | 0.7852 |
| Random Forest | 0.8279 | 0.1706 | 0.7665 |
| Logistic Reg. | 0.7219 | 0.2161 | 0.7074 |

LightGBM and XGBoost differ by only 0.0005 AUC, within the tiebreak threshold. LightGBM wins on Brier score, which matters because calibrated probabilities enter the optimizer directly: a miscalibrated classifier would systematically over- or under-discount. The AUC of 0.853 exceeds the 0.70 minimum threshold from the project proposal. The empirical waste fraction is  $\phi = 0.1727$  (17.3% of initial inventory wasted per SKU-day on average), serving as the scale constant that converts  $\rho$  into expected units wasted. Figure 1 (right panel) shows the AUC comparison across all four classifiers.

### 6.4. Strategy Comparison

Table 4 and Fig. 2 present the 60-day simulation key performance indicators across the three strategies.



**FIGURE 2: Strategy comparison across total profit (left), waste rate (center), and sell-through rate (right) over the 60-day holdout period.**

Dynamic pricing achieves a profit of \$1,999,037 versus \$419,699 under no discount, a gain of \$1,579,338 (376%). Revenue is higher (\$11.85M vs. \$10.27M), waste is lower (16.8% vs. 33.5%), and sell-through is higher (79% vs. 66%).

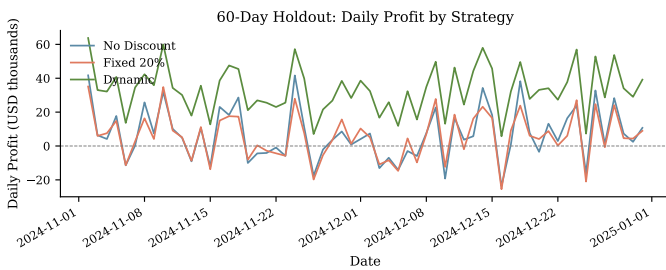
The fixed 20% strategy is the most instructive comparison. It reduces waste to 22.9%, confirming that discounting increases velocity. Yet it delivers *lower* profit than no discount (\$349,287

**TABLE 4: 60-Day Out-of-Sample Simulation KPIs by Strategy**

| Strategy    | Revenue (USD) | Profit (USD) | Waste (units) | Sell-Thru | Waste% |
|-------------|---------------|--------------|---------------|-----------|--------|
| No Discount | \$10,274,040  | \$419,699    | 567,613       | 66.0%     | 33.5%  |
| Fixed 20%   | \$10,220,438  | \$349,287    | 387,782       | 75.0%     | 22.9%  |
| Dynamic     | \$11,853,378  | \$1,999,037  | 284,554       | 79.0%     | 16.8%  |

vs. \$419,699, a deficit of \$70,413). A uniform markdown erodes margin on items that would have sold at full price regardless and generates insufficient incremental volume on low-elasticity categories to compensate. This confirms the core argument: selective discounting, concentrated on high-spoilage-risk items, is what converts velocity improvements into profit gains.

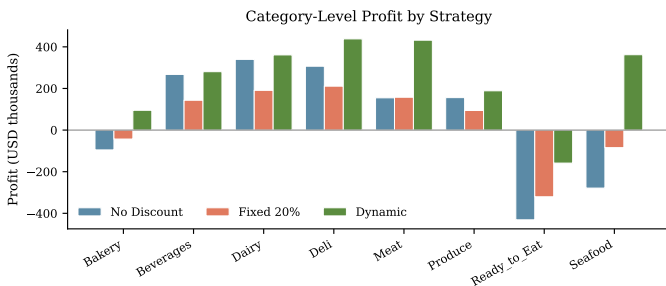
Figure 3 shows day-by-day profit trajectories. The dynamic strategy consistently dominates, and the no-discount curve shows several deeply negative days tied to spoilage events in Ready-to-Eat and Bakery.



**FIGURE 3: Daily profit trajectories for three pricing strategies over the 60-day holdout. The dashed line marks break-even.**

### 6.5. Category-Level Analysis

Figure 4 and Table 5 show per-category profit contributions.



**FIGURE 4: Per-category profit by strategy. Categories with negative profit under no-discount reveal where the dynamic strategy provides the largest gains.**

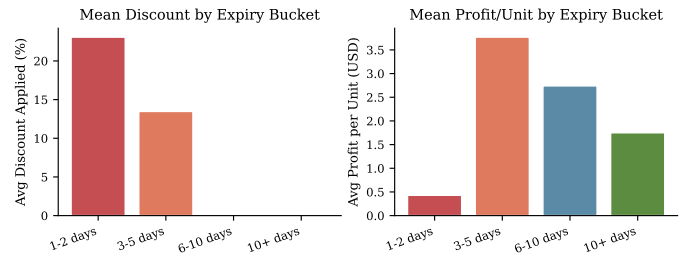
Bakery reverses a \$95,207 loss under no discount into a \$95,066 gain under dynamic pricing. Seafood turns a \$278,708 loss into a \$362,102 gain, the largest single-category swing in the simulation. Both recoveries follow the same pattern: the optimizer applies aggressive markdowns near expiry while holding full price on fresh inventory.

Ready-to-Eat reduces its loss from \$431,197 to \$158,587 under dynamic pricing, a \$272,610 improvement. Pricing alone

narrows this gap substantially; the category’s remaining deficit reflects a structural combination of thin margins and persistently high spoilage rates that calls for complementary supply-chain interventions beyond markdown policy. This direction is discussed in Section 7.

### 6.6. Near-Expiry Behavior

Table 6 and Fig. 5 show the discount and profit behavior by shelf-life bucket under the dynamic strategy.



**FIGURE 5: Mean discount (left) and mean profit per unit (right) by days-to-expiry bucket under the dynamic strategy.**

Items with 1–2 days remaining receive an average 23.1% discount, and 76.3% receive some markdown. Profit-per-unit is \$0.43, reflecting that the discount is just sufficient to move most inventory before expiry. The 3–5-day bucket is more profitable (\$3.77 per unit) because a moderate 13.5% average discount accelerates sales while preserving more margin. Items with six or more days of shelf life receive zero discount throughout the simulation. The optimizer applies markdowns exactly where spoilage risk justifies the margin sacrifice and nowhere else.

### 6.7. Sensitivity Analysis

The high-spoilage subset (days-to-expiry  $\leq 3$ , 4,021 rows) generated a waste rate of 56.9% and a total profit of  $-\$1,227,438$  under no discount. Dynamic pricing cut the waste rate to 28.9% and reduced the loss to  $-\$112,056$ , a savings of \$1,115,382. This confirms that the dynamic strategy’s advantage is largest precisely where it is most needed: imminent-expiry items.

The elasticity sensitivity experiment demonstrates optimizer robustness. Scaling the estimated category elasticities by 0.5 $\times$  or 1.5 $\times$  produces no change in which discount levels are chosen on the full holdout. Mean profit across the three scenarios varies by less than 0.5% when the analysis is restricted to non-near-expiry rows. The optimizer is insensitive to moderate mis-specification of the regular elasticity, which is a desirable property for deployment in environments where elasticity estimates carry measurement uncertainty.

The stability sample (1,000 pre-holdout rows) shows 56.3% of SKUs receiving no discount, 18.8% receiving 20%, 11.5%

**TABLE 5: Category Profit by Strategy (USD, 60-Day Holdout)**

| Category     | No Discount | Fixed 20%  | Dynamic    |
|--------------|-------------|------------|------------|
| Bakery       | −\$95,207   | −\$43,000  | \$95,066   |
| Beverages    | \$267,423   | \$143,247  | \$280,548  |
| Dairy        | \$339,715   | \$190,864  | \$361,186  |
| Deli         | \$306,861   | \$211,007  | \$438,148  |
| Meat         | \$154,891   | \$157,786  | \$431,919  |
| Produce      | \$155,922   | \$94,102   | \$188,657  |
| Ready-to-Eat | −\$431,197  | −\$320,415 | −\$158,587 |
| Seafood      | −\$278,708  | −\$84,303  | \$362,102  |

**TABLE 6: Dynamic Strategy: Discount and Profit by Expiry Bucket**

| Bucket    | $n$   | Avg Disc. | \$/unit | % Disc. |
|-----------|-------|-----------|---------|---------|
| 1–2 days  | 2,100 | 23.1%     | \$0.43  | 76.3%   |
| 3–5 days  | 1,844 | 13.5%     | \$3.77  | 55.5%   |
| 6–10 days | 1,328 | 0.0%      | \$2.74  | 0.0%    |
| 10+ days  | 1,414 | 0.0%      | \$1.75  | 0.0%    |

receiving 30%, and 9.2% receiving 40%. The 10% tier attracts only 4.2%. Most inventory is fresh enough to sell at full price; deep discounts are reserved for the high-spoilage tail.

## 6.8. Practical Recommendations

Based on the simulation findings, the following concrete recommendations apply directly to retail deployment.

*Adopt category-specific markdown policies.* Blanket discounts destroy margin on low-elasticity, low-spoilage categories such as Beverages and Dairy. The dynamic strategy raises Dairy profit from \$339,715 to \$361,186 by applying only a 2.2% mean discount and applying it selectively, not uniformly.

*Concentrate discount aggressiveness within five days of expiry.* Items with 1–2 days remaining warrant discounts averaging 23%, while items with six or more days warrant no discount at all. Retailers should configure alert thresholds and trigger automated markdowns at this boundary rather than applying time-insensitive rules.

*Track Ready-to-Eat margin structure separately.* This category benefits substantially from dynamic pricing (\$272,610 loss reduction) but does not cross into profitability. Its margin profile and spoilage rate should be monitored as a distinct business problem, with ordering quantities adjusted as a complementary lever.

*Recompute the empirical waste fraction periodically.* The  $\phi = 0.1727$  calibration constant converts spoilage probability into expected wasted units. Seasonal variation in consumer behavior and supply conditions will shift this value; quarterly recalibration keeps the optimizer’s waste estimates aligned with observed operations.

*Treat spoilage probability calibration as a first-class output.* LightGBM was selected partly because its Brier score (0.1535) reflects well-calibrated probability estimates. Replacing the spoilage classifier without verifying calibration on the new model would silently degrade optimizer performance.

## 7. FUTURE WORK

The results here open three natural extensions.

*Estimating near-expiry elasticity from data.* The near-expiry elasticity  $\epsilon_{ne} = -2.0$  is derived from published literature ranges (−1.5 to −3.0). The sensitivity analysis confirms that this parameter dominates optimizer behavior in the regime where markdowns are most active. Estimating it directly from transaction data – for example, by fitting a separate log-log regression on SKUs observed within five days of expiry – would replace a literature-calibrated assumption with an evidence-based one, potentially further improving profit in the near-expiry buckets.

*Joint pricing and ordering optimization.* The current framework takes inventory levels as given and optimizes only the markdown decision. Categories such as Ready-to-Eat, where the combination of thin margins and high spoilage creates structural losses that pricing alone cannot close, would benefit from co-optimizing the initial order quantity alongside the markdown schedule. Extending the profit objective of Eq. (2) to include ordering costs and incorporating demand uncertainty into the order decision is a natural next step grounded in the inventory-theoretic literature [1, 2].

*Ensemble forecasting and learned pricing policies.* Prophet [9] offers explicit seasonal decomposition that may benefit categories with strong annual cycles, and an ensemble combining LightGBM with a decomposable model could yield better directional accuracy for those categories. Looking further ahead, the reinforcement learning framework of Liu et al. [5] provides a path toward jointly optimizing pricing and forecasting rather than treating them as sequential stages, which could capture non-stationary demand dynamics that the current fixed-elasticity model does not model.

## 8. CONCLUSION

This project presented a validated five-stage machine learning pipeline for dynamic pricing of perishable grocery inventory. LightGBM achieved a validation RMSE of 858 units and directional accuracy of 62.3% for demand forecasting, and a ROC-AUC of 0.853 for spoilage risk classification. Log-log panel regressions on the Dunnhumby dataset produced inelastic price elasticities ranging from −0.055 for Meat to −0.245 for Pastry, confirming that demand heterogeneity across categories makes uniform markdown policies suboptimal.

The 60-day out-of-sample simulation delivered the central result: dynamic pricing raised profit from \$419,699 to \$1,999,037, a gain of \$1,579,338 (376%), while simultaneously

cutting the waste rate from 33.5% to 16.8% and increasing sell-through from 66% to 79%. Categories that were unprofitable under static pricing – Bakery (\$95K gain) and Seafood (\$362K gain) – turned profitable under the optimizer. The sensitivity analyses confirm the framework is robust to elasticity mis-specification and that its largest benefits concentrate exactly where they are needed most: imminent-expiry inventory.

The pipeline is fully operational. All models are serialized, all outputs are versioned, and a Streamlit dashboard supports live what-if exploration without retraining. The next development cycle, described in Section 7, focuses on estimating the near-expiry elasticity empirically and extending the optimizer to cover joint ordering decisions, which would address the structural constraints that pricing alone cannot close. The framework is designed to grow incrementally: each extension slots into the existing modular pipeline without requiring a ground-up rebuild.

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